Better approximation to the susie-love model

1 TO-DO

Understand the relationship between

1. PIP only depends on x and the diagonal of R.
2. $p(x_{-j}|x_j, \beta, \gamma = j)$ does not depend on β.

Look Yuxin's thesis and read the SI of susie-rss paper more carefully.

Why does having β^2 in the variance make sense? Of course it follows from the IW or Wi model, but intuitively why does it make sense?

Is there another model that arises from the conditional distribution:

$$p(x_{-j} \mid x_j, \beta, \gamma = j) = \mathcal{N}\left(x_j \bar{R}_{j,-j}, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0 + 3}\right) S_j\right)? \tag{1}$$

2 Some new models with better approximation to susie-love

Recall our original SER model from what is called 'susie-love':

$$\begin{cases} x|R, \beta, \gamma \sim \mathcal{N}(\beta R_\gamma, R/N) \\ R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1). \end{cases} \tag{2}$$

Note that the latent matrix R appears in both mean and variance of the likelihood in (2). This model is very natural for unknown in-sample LD R, but is hard to make inferences. Last time we approximated R in the variance to be $\bar{R}$, which makes the inference much easier but might eventually lead to the loss of the "irrelevance of the null SNPs" property. In principle, it makes sense that when $L = 1$, PIP of SNP j calculation should only depend on x_j but not on other x_k and $\bar{R}_{jk}$ ("irrelevance of the null SNPs" property for SER).

I think this original model is actually feasible for inference. Consider the following equivalent way to represent it.

2.1 Model A (equivalent to the original SER-love model)

Let's write down $p(x|R, \beta, \gamma = 1)$, other cases of γ are similar. Let

$$d_1 = R_{11} \in \mathbb{R}_+, \quad z_1 = \frac{R_{-1,1}}{d_1} \in \mathbb{R}^{J-1},$$

and

$$S_1 = R_{-1,-1} - d_1 z_1 z_1^\top$$

be the Schur's complement of R_{11} in R. Hence, the latent R is represented as

$$R = \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}. \tag{3}$$

Similarly, denote

$$\bar{S}_1 = \bar{R}_{-1,-1} - \bar{R}_{-1,1} \bar{R}_{-1,1}^\top$$

be the Schur's complement of $\bar{R}_{11}$ in $\bar{R}$, and $\mu_1 = \bar{R}_{-1,1}$, we can write

$$\bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix}. \tag{4}$$

Properties of IW distribution (together with the fact that $\bar{R}_{11} = 1$) imply, using the shape–scale parameterization for the inverse-gamma distribution,

$$\begin{cases} d_1 \sim \mathrm{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\bar{R}_{-1,1}, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \end{cases} \tag{5}$$

The original SER version of Susie-love (with $\gamma = 1$; similar for other γ) becomes:

$$\begin{cases} x | \beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}\right) \\ d_1 \sim \mathrm{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \tag{6}$$

Deriving $p(x \mid \gamma = 1)$. Write

$$x = \begin{pmatrix} x_1 \\ x_{-1} \end{pmatrix}.$$

Step 1: likelihood conditional on all latent variables. From the block normal distribution in (6),

$$p(x \mid \gamma = 1, \beta, d_1, z_1, S_1) = \mathcal{N}_J\left(x \mid \beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}\right).$$

Equivalently, using the conditional distribution of a block normal,

$$p(x \mid \gamma = 1, \beta, d_1, z_1, S_1) = \mathcal{N}\left(x_1 \mid \beta d_1, \frac{d_1}{N}\right) \mathcal{N}_{J-1}\left(x_{-1} \mid x_1 z_1, \frac{S_1}{N}\right).$$

The key simplification is that, **after conditioning on x_1, the conditional mean of x_{-1} is $x_1 z_1$ and no longer contains β or d_1** (relevant to our earlier discussion).

Step 2: marginalize over $z_1 \mid S_1$. Since

$$z_1 \mid S_1 \sim \mathcal{N}_{J-1}\left(\mu_1, \frac{S_1}{\nu_0}\right)$$

and

$$x_{-1} \mid x_1, z_1, S_1 \sim \mathcal{N}_{J-1}\left(x_1 z_1, \frac{S_1}{N}\right),$$

we get

$$x_{-1} \mid x_1, S_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1\mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) S_1\right). \tag{7}$$

Therefore

$$p(x \mid \gamma = 1, \beta, d_1, S_1) = \mathcal{N}\left(x_1 \mid \beta d_1, \frac{d_1}{N}\right) \mathcal{N}_{J-1}\left(x_{-1} \mid x_1\mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) S_1\right).$$

What I found so interesting is that the conditional distribution $p(x_{-1} \mid x_1, S_1, \gamma = 1)$ resembles what was suggested by Matthew last time (cf. (1)).

Step 3: marginalize over S_1. Because $S_1 \sim \mathcal{IW}_{J-1}(\nu_0\bar{S}_1, \nu_0 + J + 1)$, integrating S_1 gives a multivariate t distribution with $\nu_0 + 3$ degrees of freedom:

$$x_{-1} \mid x_1, \gamma = 1 \sim t_{J-1,\nu_0+3}\left(x_1\mu_1, \frac{\nu_0}{\nu_0+3}\left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right)\bar{S}_1\right),$$

where the second argument is the multivariate t scale matrix. Thus

$$p(x \mid \gamma = 1, \beta, d_1) = \mathcal{N}\left(x_1 \mid \beta d_1, \frac{d_1}{N}\right) t_{J-1,\nu_0+3}\left(x_{-1} \mid x_1\mu_1, \frac{\nu_0}{\nu_0+3}\left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right)\bar{S}_1\right).$$

Equivalently, the t factor is proportional to

$$|\bar{S}_1|^{-1/2}\left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right)^{-(J-1)/2}\left[1 + \frac{(x_{-1} - x_1\mu_1)^\top(\nu_0\bar{S}_1)^{-1}(x_{-1} - x_1\mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}}\right]^{-(\nu_0+J+2)/2}.$$

Using the block inverse formula (notice the block form of $\bar{R}$ (4)),

$$x^\top\bar{R}^{-1}x = x_1^2 + (x_{-1} - x_1\mu_1)^\top\bar{S}_1^{-1}(x_{-1} - x_1\mu_1),$$

so

$$1 + \frac{(x_{-1} - x_1\mu_1)^\top(\nu_0\bar{S}_1)^{-1}(x_{-1} - x_1\mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}} = \frac{x^\top\bar{R}^{-1}x + \nu_0/N}{x_1^2 + \nu_0/N}.$$

Therefore

$$T_1(x) \propto |\bar{S}_1|^{-1/2}\left(x_1^2 + \frac{\nu_0}{N}\right)^{(\nu_0+3)/2}\left(x^\top\bar{R}^{-1}x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2}. \tag{8}$$

When comparing different values of γ, the final factor is common across SNPs. Also, since $\bar{R}_{11} = 1$, $|\bar{R}| = |\bar{S}_1|$; analogously, $|\bar{S}_j| = |\bar{R}|$ for any SNP j with $\bar{R}_{jj} = 1$.

Step 4: marginalize over β and d_1. The preceding display separates the problem into an $x_{-1} \mid x_1$ factor and an x_1 factor. Define

$$T_1(x) = t_{J-1,\nu_0+3}\left(x_{-1} \mid x_1\mu_1,\, \frac{\nu_0}{\nu_0+3}\left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right)\bar{S}_1\right).$$

If we integrate over β, then

$$x_1 \mid d_1, \gamma = 1 \sim \mathcal{N}\left(0, \frac{d_1}{N} + d_1^2\sigma_0^2\right),$$

and hence (finally integrating over d_1):

$$\boxed{p(x \mid \gamma = 1) = T_1(x)\int_0^\infty \mathcal{N}\left(x_1 \mid 0, \frac{d_1}{N} + d_1^2\sigma_0^2\right)\mathrm{IG}\left(d_1; \frac{\nu_0+2}{2}, \frac{\nu_0}{2}\right) dd_1}.$$

This is a one-dimensional integral over $d_1 > 0$. I do not see a standard closed form for it because the variance is $d_1/N + d_1^2\sigma_0^2$. Replacing T_1 with its expression in (8),

$$\begin{aligned} p(x \mid \gamma = 1) \propto |\bar{R}|^{-1/2}\left(x_1^2 + \frac{\nu_0}{N}\right)^{(\nu_0+3)/2}\left(x^\top \bar{R}^{-1}x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2} \\ \times \int_0^\infty \mathcal{N}\left(x_1 \mid 0, \frac{d_1}{N} + d_1^2\sigma_0^2\right) d\,\mathrm{IG}(d_1) \end{aligned}$$

Since $|\bar{R}|$ and $x^\top \bar{R}^{-1}x$ do not depend on which SNP is chosen, both factors cancel in the PIP normalization. Also,

$$\left(x_j^2 + \frac{\nu_0}{N}\right)^{(\nu_0+3)/2} = \left(\frac{\nu_0}{N}\right)^{(\nu_0+3)/2}\left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2},$$

and the first factor is common across SNPs. Hence,

$$\boxed{p(\gamma = j \mid x) \propto \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2}\int_0^\infty \mathcal{N}\left(x_j \mid 0, \frac{d_j}{N} + d_j^2\sigma_0^2\right) d\,\mathrm{IG}(d_j)}.$$

Conclusion. In the original SER-love model, the marginal likelihood depends on $x^\top \bar{R}^{-1}x$, but this dependence is common across SNPs and cancels when computing the PIP. The resulting PIP weight depends on SNP j only through x_j.

2.2 Model B

Simplify $d_1 \equiv 1$ to avoid intractable integral (and we know the diagonal of the in-sample LD matrix is 1 anyway), we have the following model (model B):

$$\begin{cases} x|\beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta\begin{pmatrix}1\\ z_1\end{pmatrix}, \frac{1}{N}\begin{pmatrix}1 & z_1^\top \\ z_1 & S_1 + z_1z_1^\top\end{pmatrix}\right) \\ z_1|S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0\bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \tag{9}$$

This model leads to

$$\boxed{p(x \mid \gamma = 1) \propto |\bar{R}|^{-1/2} \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1} x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2} \mathcal{N}\left(x_1 \mid 0, \frac{1}{N} + \sigma_0^2\right)}.$$

Since $|\bar{R}|$ and $x^\top \bar{R}^{-1} x$ do not depend on which SNP is chosen, the PIP weights simplify further. Dropping the common factors gives

$$\boxed{p(\gamma = j \mid x) \propto \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \mathcal{N}\left(x_j \mid 0, \frac{1}{N} + \sigma_0^2\right)}. \tag{10}$$

Compare to SER-RSS's PIP. Recall SER-RSS's PIP:

$$\tilde{p}(\gamma = j \mid x) \propto \exp\left(\frac{N\sigma_0^2}{N\sigma_0^2 + 1} \frac{Nx_j^2}{2}\right) \tag{11}$$

We now show that (10) tends to (11) as $\nu_0 \to \infty$. Since

$$\left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} = \left(1 + \frac{2}{\nu_0 + 3} \times \frac{\nu_0 + 3}{\nu_0} \frac{Nx_j^2}{2}\right)^{(\nu_0+3)/2} \asymp \exp\left\{\frac{\nu_0 + 3}{\nu_0} \frac{Nx_j^2}{2}\right\}$$

Also,

$$\mathcal{N}\left(x_j \mid 0, \frac{1}{N} + \sigma_0^2\right) \propto \exp\left\{-\frac{x_j^2}{2(1/N + \sigma_0^2)}\right\} = \exp\left\{-\frac{1}{1 + N\sigma_0^2} \frac{Nx_j^2}{2}\right\}.$$

Therefore the limiting weight in (10) is asymptotically (as ν_0 gets large)

$$\exp\left\{\left(\frac{\nu_0 + 3}{\nu_0} - \frac{1}{N\sigma_0^2 + 1}\right) \frac{Nx_j^2}{2}\right\} = \exp\left\{\left(\frac{N\sigma_0^2}{N\sigma_0^2 + 1} - \frac{1}{\nu_0 + 3}\right) \frac{Nx_j^2}{2}\right\} \approx \exp\left\{\frac{N\sigma_0^2}{N\sigma_0^2 + 1} \frac{Nx_j^2}{2}\right\}$$

which is exactly the SER-RSS PIP weight in (11). Hence, for finite ν_0, PIP under SER-love model (B) is slightly more diffused than susie-rss.

Inference under model B (9). For a general SNP j, write

$$\mu_j = \bar{R}_{-j,j}, \qquad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top, \qquad \kappa_j = \nu_0 + Nx_j^2.$$

Under model B, conditional on $\gamma = j$, the likelihood factorizes as

$$p(x \mid \beta, z_j, S_j, \gamma = j) = \mathcal{N}\left(x_j \mid \beta, \frac{1}{N}\right) \mathcal{N}_{J-1}\left(x_{-j} \mid x_j z_j, \frac{S_j}{N}\right).$$

Thus β is conditionally independent of (z_j, S_j) given x and $\gamma = j$, and

$$p(\beta, z_j, S_j \mid x, \gamma = j) = p(\beta \mid x_j, \gamma = j)\, p(z_j, S_j \mid x, \gamma = j).$$

- The posterior for β is the usual normal-normal update:

$$\boxed{\beta \mid x, \gamma = j \sim \mathcal{N}\left(\frac{N}{N + \sigma_0^{-2}} x_j,\ \frac{1}{N + \sigma_0^{-2}}\right).}$$

- For the LD-column variables, first condition on S_j. Combining

$$z_j \mid S_j \sim \mathcal{N}_{J-1}\left(\mu_j, \frac{S_j}{\nu_0}\right), \qquad x_{-j} \mid x_j, z_j, S_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j z_j, \frac{S_j}{N}\right)$$

gives

$$\boxed{z_j \mid S_j, x, \gamma = j \sim \mathcal{N}_{J-1}\left(\frac{\nu_0 \mu_j + N x_j x_{-j}}{\nu_0 + N x_j^2}, \frac{S_j}{\nu_0 + N x_j^2}\right).}$$

- Integrating out z_j gives

$$x_{-j} \mid x_j, S_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j \mu_j, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right) S_j\right).$$

Therefore the posterior for S_j is inverse-Wishart:

$$\boxed{S_j \mid x, \gamma = j \sim \mathcal{IW}\left(\nu_0 \bar{S}_j + \frac{N\nu_0}{\nu_0 + N x_j^2}(x_{-j} - x_j \mu_j)(x_{-j} - x_j \mu_j)^\top,\ \nu_0 + J + 2\right).}$$

Equivalently, the joint posterior of (z_j, S_j) can be represented by first drawing S_j from this inverse-Wishart distribution and then drawing z_j from the conditional normal distribution above. Finally, the full posterior over γ uses the PIP weights in (10), normalized over $j = 1, \ldots, J$.

Closed-form ELBO. Because the conditional posteriors above are exact, the optimized ELBO for model B is the exact marginal log likelihood. Let

$$V_\beta = \frac{1}{N} + \sigma_0^2, \qquad q = x^\top \bar{R}^{-1} x.$$

For a uniform prior on γ, the optimized ELBO is

$$\mathcal{L}(\nu_0, \sigma_0^2) = \log\left\{\frac{1}{J}\sum_{j=1}^{J} \exp(\ell_j)\right\},$$

where the component log marginal likelihood can be written as

$$\begin{aligned}\ell_j = & -\frac{1}{2}\log(2\pi V_\beta) - \frac{x_j^2}{2V_\beta} + \log\Gamma\left(\frac{\nu_0 + J + 2}{2}\right) - \log\Gamma\left(\frac{\nu_0 + 3}{2}\right) \\ & - \frac{J-1}{2}\log\pi - \frac{1}{2}\log|\bar{S}_j| + \frac{\nu_0 + 3}{2}\log\left(x_j^2 + \frac{\nu_0}{N}\right) - \frac{\nu_0 + J + 2}{2}\log\left(q + \frac{\nu_0}{N}\right).\end{aligned}$$

This is equivalent to the marginal likelihood derived above, but unlike the PIP formula it keeps the terms that are common across SNPs. Those common terms do not affect $\pi_j = p(\gamma = j \mid x)$ for fixed hyperparameters, but they do matter when learning ν_0.

Empirical Bayes learning of ν_0 and σ_0^2. The empirical Bayes estimates can be obtained by maximizing the closed-form objective

$$(\hat{\nu}_0, \hat{\sigma}_0^2) = \arg\max_{\nu_0>0,\ \sigma_0^2\geq 0} \mathcal{L}(\nu_0, \sigma_0^2).$$

After choosing $(\hat{\nu}_0, \hat{\sigma}_0^2)$, the posterior inclusion probabilities are

$$\pi_j = \left.\frac{\exp(\ell_j)}{\sum_{k=1}^J \exp(\ell_k)}\right|_{\nu_0=\hat{\nu}_0,\ \sigma_0^2=\hat{\sigma}_0^2}.$$

Equivalently, for computing PIPs at fixed hyperparameters one may use the simplified weights in (10); for empirical Bayes optimization one should use $\mathcal{L}(\nu_0, \sigma_0^2)$ above, because it retains the ν_0-dependent normalizing constants and the common $\log(q + \nu_0/N)$ term.

EM view of empirical Bayes. Another way to use the fact that the posterior is exact is to write the usual ELBO identity. Let

$$\theta = (\gamma, \beta, z_\gamma, S_\gamma), \qquad q(\theta) = p(\theta \mid x; \nu_0^{\text{old}}, \sigma_{0,\text{old}}^2).$$

Strictly speaking,

$$\log p(x; \nu_0, \sigma_0^2) \geq \mathbb{E}_q\{\log p(x \mid \theta)\} - \mathrm{KL}\{q(\theta) \,\|\, p(\theta; \nu_0, \sigma_0^2)\},$$

with equality when q is the posterior under the same hyperparameters. In an EM algorithm, the E-step fixes q at the current hyperparameters and the M-step maximizes

$$Q(\nu_0, \sigma_0^2) = \mathbb{E}_q\{\log p(\theta; \nu_0, \sigma_0^2)\},$$

because the likelihood term does not involve ν_0 or σ_0^2 in model B.

This gives a closed-form update for σ_0^2. If

$$\pi_j = q(\gamma = j), \qquad m_{\beta j} = \frac{N}{N + \sigma_{0,\text{old}}^{-2}} x_j, \qquad v_\beta = \frac{1}{N + \sigma_{0,\text{old}}^{-2}},$$

then

$$\boxed{\sigma_{0,\text{new}}^2 = \sum_{j=1}^J \pi_j \left(m_{\beta j}^2 + v_\beta\right).}$$

For ν_0, the same EM view reduces the update to a one-dimensional optimization, but not to a closed-form expression. Let $p = J - 1$ and write the current posterior of S_j as

$$S_j \mid x, \gamma = j \sim \mathcal{IW}_p(\Psi_j, m_j), \qquad m_j = \nu_0^{\text{old}} + J + 2,$$

where

$$\Psi_j = \nu_0^{\text{old}} \bar{S}_j + \frac{N\nu_0^{\text{old}}}{\nu_0^{\text{old}} + N x_j^2}(x_{-j} - x_j\mu_j)(x_{-j} - x_j\mu_j)^\top.$$

Also let

$$\bar{z}_j = \frac{\nu_0^{\text{old}}\mu_j + N x_j x_{-j}}{\nu_0^{\text{old}} + N x_j^2}, \qquad \kappa_j^{\text{old}} = \nu_0^{\text{old}} + N x_j^2.$$

These quantities can be evaluated without forming or inverting the $(J-1)\times(J-1)$ matrices Ψ_j. Let

$$\nu_\star = \nu_0^{\text{old}}, \qquad q_R = x^\top \bar{R}^{-1} x, \qquad \delta_j = x_{-j} - x_j \mu_j.$$

Then

$$\Psi_j = \nu_\star \bar{S}_j + \frac{N\nu_\star}{\kappa_j^{\text{old}}} \delta_j \delta_j^\top.$$

By the Woodbury identity,

$$\Psi_j^{-1} = \frac{1}{\nu_\star} \bar{S}_j^{-1} - \frac{N}{\nu_\star(\nu_\star + N q_R)} \bar{S}_j^{-1} \delta_j \delta_j^\top \bar{S}_j^{-1},$$

because

$$1 + \frac{N}{\kappa_j^{\text{old}}}(q_R - x_j^2) = \frac{\nu_\star + N q_R}{\kappa_j^{\text{old}}}.$$

The determinant lemma gives

$$\log|\Psi_j| = (J-1)\log\nu_\star + \log|\bar{S}_j| + \log\left(\frac{\nu_\star + N q_R}{\kappa_j^{\text{old}}}\right).$$

Since $\bar{R}_{jj} = 1$, $\log|\bar{S}_j| = \log|\bar{R}|$, so this is also common to compute once.

The needed posterior expectations are

$$A_j = \mathbb{E}_q\{\operatorname{tr}(\bar{S}_j S_j^{-1})\} = m_j \left\{ \frac{J-1}{\nu_\star} - \frac{N(q_R - x_j^2)}{\nu_\star(\nu_\star + N q_R)} \right\},$$

$$B_j = \mathbb{E}_q\{(z_j - \mu_j)^\top S_j^{-1}(z_j - \mu_j)\} = m_j \frac{N^2 x_j^2 (q_R - x_j^2)}{\nu_\star \kappa_j^{\text{old}} (\nu_\star + N q_R)} + \frac{p}{\kappa_j^{\text{old}}},$$

and

$$C_j = \mathbb{E}_q\{\log|S_j|\} = \ (J-1)\log\nu_\star + \log|\bar{R}| + \log\left(\frac{\nu_\star + N q_R}{\kappa_j^{\text{old}}}\right) - \sum_{r=1}^{p} \psi\left(\frac{m_j + 1 - r}{2}\right) - p\log 2.$$

Thus all SNP-specific parts of A_j, B_j, C_j depend on j only through x_j^2 and $q_R - x_j^2$. In implementation, one can compute q_R and $\log|\bar{R}|$ once, then evaluate the remaining terms with scalar operations for each SNP. Up to constants that do not depend on ν_0, the EM objective for ν_0 is

$$Q_\nu(\nu_0) = \sum_{j=1}^{J} \pi_j \Bigg[\frac{p}{2}\log\nu_0 + \frac{\nu_0 + J + 1}{2}\log|\nu_0 \bar{S}_j| - \frac{(\nu_0 + J + 1)p}{2}\log 2$$
$$- \log\Gamma_p\left(\frac{\nu_0 + J + 1}{2}\right) - \frac{\nu_0 + 2J + 2}{2} C_j - \frac{\nu_0}{2}(A_j + B_j) \Bigg].$$

Thus

$$\boxed{\nu_{0,\text{new}} = \arg\max_{\nu_0 > 0} Q_\nu(\nu_0).}$$

So the exact-posterior ELBO gives a particularly nice update for σ_0^2, and a stable scalar optimization for ν_0. For ν_0, the closed-form marginal likelihood $\mathcal{L}(\nu_0, \sigma_0^2)$ above is often even simpler to optimize directly, but the EM form is useful for deriving a monotone coordinate-ascent algorithm.

2.3 Other representations of Model B

Recall Model B:

$$\begin{cases} x|\beta,\gamma=1,z_1,S_1 \sim \mathcal{N}\left(\beta\begin{pmatrix}1\\ z_1\end{pmatrix}, \frac{1}{N}\begin{pmatrix}1 & z_1^\top\\ z_1 & S_1+z_1z_1^\top\end{pmatrix}\right)\\ z_1|S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right)\\ S_1 \sim \mathcal{IW}(\nu_0\bar{S}_1, \nu_0+J+1)\\ \beta \sim \mathcal{N}(0,\sigma_0^2)\end{cases} \tag{12}$$

or equivalently,

$$\begin{cases} x_1 \mid \beta,\gamma=1 \sim \mathcal{N}\left(\beta, \frac{1}{N}\right),\\ x_{-1} \mid x_1, z_1, \gamma=1 \sim \mathcal{N}_{J-1}\left(x_1z_1, \frac{1}{N}S_1\right),\\ z_1|S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right),\\ S_1 \sim \mathcal{IW}(\nu_0\bar{S}_1, \nu_0+J+1),\\ \beta \sim \mathcal{N}(0,\sigma_0^2).\end{cases} \tag{13}$$

Here are three other equivalent representations of model B for $\gamma = 1$. Write

$$\mu_1 = \bar{R}_{-1,1}, \qquad \bar{S}_1 = \bar{R}_{-1,-1} - \mu_1\mu_1^\top.$$

1. After integrating out $z_1 \mid S_1$.

$$\begin{cases} x_1 \mid \beta,\gamma=1 \sim \mathcal{N}\left(\beta, \frac{1}{N}\right),\\ x_{-1} \mid x_1, S_1, \gamma=1 \sim \mathcal{N}_{J-1}\left(x_1\mu_1, \left(\frac{1}{N}+\frac{x_1^2}{\nu_0}\right)S_1\right),\\ S_1 \sim \mathcal{IW}(\nu_0\bar{S}_1, \nu_0+J+1),\\ \beta \sim \mathcal{N}(0,\sigma_0^2).\end{cases} \tag{14}$$

2. After further integrating out S_1.

$$\begin{cases} x_1 \mid \beta,\gamma=1 \sim \mathcal{N}\left(\beta, \frac{1}{N}\right),\\ x_{-1} \mid x_1, \gamma=1 \sim t_{J-1,\nu_0+3}\left(x_1\mu_1, \frac{\nu_0}{\nu_0+3}\left(\frac{1}{N}+\frac{x_1^2}{\nu_0}\right)\bar{S}_1\right),\\ \beta \sim \mathcal{N}(0,\sigma_0^2).\end{cases} \tag{15}$$

3. Using the normal–Gamma mixture for the t distribution. With shape–rate parameterization, Conditional on ω_1:

$$
\begin{cases}
x_1 \mid \beta, \gamma = 1 \sim \mathcal{N}\left(\beta, \dfrac{1}{N}\right), \\
x_{-1} \mid x_1, \omega_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1 \mu_1, \dfrac{1}{\omega_1}\left(\dfrac{1}{N} + \dfrac{x_1^2}{\nu_0}\right)\bar{S}_1\right), \\
\omega_1 \sim \text{Gamma}\left(\dfrac{\nu_0+3}{2}, \dfrac{\nu_0}{2}\right), \\
\beta \sim \mathcal{N}(0, \sigma_0^2).
\end{cases}
\tag{16}
$$

2.4 A shared ω model

shared-ω model

$$
x \mid \beta, \gamma = j, z_j, \omega_j \sim N_J\left(\beta u_j(z_j), \frac{\bar{R}}{N\omega_j}\right),
$$

$$
z_j \mid \omega_j \sim N_{J-1}\left(\mu_j, \frac{\bar{S}_j}{\nu_0 \omega_j}\right), \qquad \omega_j \sim \text{Gamma}\left(\frac{\nu_0+3}{2}, \frac{\nu_0}{2}\right).
$$

This model is somewhat between model B and SER-relax. PIP in this model does not satisfy the "irrelevance of null SNPs" property but it approximates Model B way better than SER-relax (whose PIP is more diffuse). It might be easier to extend to a susie-like model as well?

3 Rewrite the model

Model A and B are very nice but they are hard to extend to a susie-like model due to the randomness in noise of x.

My goal now is to introduce a few latent variables to achieve the noise like $\mathcal{N}(0, \bar{R}/N)$ for the model.

Let's start with the following latent-noise representation for $\gamma = 1$:

$$
\begin{cases}
x = \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix} + z_0 \\
z_0 \mid R \sim \mathcal{N}(0, R/N) \\
z_1 = \dfrac{R_{-1,1}}{R_{11}} \\
R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1)
\end{cases}
\tag{17}
$$

First, we note that when marginalizing out R, both z_0 and z_1 are multivariate-t distributed. For the z_0 part, the useful normal–Gamma representation is

$$
z_0 = \frac{\bar{z}_0}{\sqrt{\omega_0}}, \quad \bar{z}_0 \sim \mathcal{N}\left(0, \frac{\bar{R}}{N}\right), \quad \omega_0 \sim \text{Gamma}\left(\frac{\nu_0+2}{2}, \frac{\nu_0}{2}\right),
\tag{18}
$$

using the shape–rate parameterization for the Gamma distribution. The square root is important: conditional on ω_0, z_0 has covariance $\bar{R}/(N\omega_0)$, and integrating over ω_0 gives the same marginal distribution as integrating R out of $z_0 \mid R$.

With this representation the observation equation becomes

$$x = \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix} + \frac{\bar{z}_0}{\sqrt{\omega_0}},$$

or equivalently

$$\sqrt{\omega_0} x = \sqrt{\omega_0} \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix} + \bar{z}_0, \qquad \bar{z}_0 \sim \mathcal{N}\left(0, \frac{\bar{R}}{N}\right). \tag{19}$$

Thus, conditional on ω_0, the likelihood can be written with the fixed noise covariance $\bar{R}/N$:

$$p(x \mid \beta, z_1, \omega_0) = \omega_0^{J/2} \mathcal{N}_J \left(\sqrt{\omega_0} x \,\middle|\, \sqrt{\omega_0} \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix}, \frac{\bar{R}}{N} \right).$$

Now we need the conditional distribution of z_1 after introducing $\bar{z}_0$. Since

$$R \mid z_0 \sim \mathcal{IW}\left(\nu_0 \bar{R} + N z_0 z_0^\top,\, \nu_0 + J + 2\right),$$

we can write the updated inverse-Wishart scale matrix in terms of $(\bar{z}_0, \omega_0)$ as

$$\Psi(\bar{z}_0, \omega_0) = \nu_0 \bar{R} + \frac{N}{\omega_0} \bar{z}_0 \bar{z}_0^\top.$$

Partition

$$\bar{z}_0 = \begin{pmatrix} \bar{z}_{01} \\ \bar{z}_{0,-1} \end{pmatrix}, \qquad \delta_1 = \bar{z}_{0,-1} - \bar{z}_{01} \mu_1,$$

and define

$$\kappa_0 = \nu_0 + \frac{N}{\omega_0} \bar{z}_{01}^2.$$

The relevant blocks of Ψ are

$$\Psi_{11} = \kappa_0, \qquad \Psi_{-1,1} = \nu_0 \mu_1 + \frac{N}{\omega_0} \bar{z}_{01} \bar{z}_{0,-1}.$$

Therefore the conditional mean for the LD-column variable is

$$\mu_1^\Psi = \frac{\Psi_{-1,1}}{\Psi_{11}} = \mu_1 + \frac{N \bar{z}_{01}}{\omega_0 \kappa_0} \delta_1. \tag{20}$$

The Schur complement of Ψ_{11} in Ψ is

$$S_1^\Psi = \Psi_{-1,-1} - \frac{\Psi_{-1,1} \Psi_{1,-1}}{\Psi_{11}} = \nu_0 \bar{S}_1 + \frac{\nu_0 N}{\omega_0 \kappa_0} \delta_1 \delta_1^\top. \tag{21}$$

By the inverse-Wishart block decomposition,

$$S_1 \mid \bar{z}_0, \omega_0 \sim \mathcal{IW}\left(S_1^\Psi, \nu_0 + J + 2\right), \tag{22}$$

$$z_1 \mid S_1, \bar{z}_0, \omega_0 \sim \mathcal{N}_{J-1}\left(\mu_1^\Psi, \frac{S_1}{\kappa_0}\right). \tag{23}$$

Equivalently, after integrating out S_1,

$$z_1 \mid \bar{z}_0, \omega_0 \sim t_{J-1,\nu_0+4}\left(\mu_1^\Psi,\, \frac{S_1^\Psi}{\kappa_0(\nu_0+4)}\right),$$

where the second argument is the multivariate-t scale matrix.

Introduce the normal–Gamma representation of this t distribution:

$$\omega_1 \sim \text{Gamma}\left(\frac{\nu_0+4}{2}, \frac{\nu_0+4}{2}\right), \tag{24}$$

$$z_1 \mid \bar{z}_0, \omega_0, \omega_1 \sim \mathcal{N}_{J-1}\left(\mu_1^\Psi,\, \frac{S_1^\Psi}{\omega_1\kappa_0(\nu_0+4)}\right). \tag{25}$$

The mean shift in (20) can be separated from the baseline LD column by writing

$$z_1 = \mu_1 + \frac{N\bar{z}_{01}}{\omega_0\kappa_0}\delta_1 + \varepsilon_1, \qquad \varepsilon_1 \mid \bar{z}_0, \omega_0, \omega_1 \sim \mathcal{N}_{J-1}\left(0, \frac{S_1^\Psi}{\omega_1\kappa_0(\nu_0+4)}\right).$$

Hence the observation equation can be rewritten as

$$x = \beta\begin{pmatrix}1\\ \mu_1\end{pmatrix} + \frac{\bar{z}_0}{\sqrt{\omega_0}} + \beta\begin{pmatrix}0\\ \frac{N\bar{z}_{01}}{\omega_0\kappa_0}\delta_1\end{pmatrix} + \beta\begin{pmatrix}0\\ \varepsilon_1\end{pmatrix}. \tag{26}$$

Thus the δ_1 part of the conditional mean of z_1 has become an explicit additive noise term in the model for x.

We can further extract the rank-one part of S_1^Ψ. Let

$$\xi_1 \sim \mathcal{N}_{J-1}(0, \nu_0\bar{S}_1), \qquad \zeta_1 \sim \mathcal{N}(0,1),$$

independently conditional on $(\bar{z}_0, \omega_0, \omega_1)$. Then

$$\xi_1 + \sqrt{\frac{\nu_0 N}{\omega_0\kappa_0}}\,\zeta_1\delta_1 \sim \mathcal{N}_{J-1}(0, S_1^\Psi),$$

and therefore

$$\varepsilon_1 = \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\left\{\xi_1 + \sqrt{\frac{\nu_0 N}{\omega_0\kappa_0}}\,\zeta_1\delta_1\right\}.$$

Plugging this into (26) gives the fully extracted form

$$\begin{aligned} x = {}& \beta\begin{pmatrix}1\\ \mu_1\end{pmatrix} + \frac{\bar{z}_0}{\sqrt{\omega_0}} + \beta\begin{pmatrix}0\\ \frac{N\bar{z}_{01}}{\omega_0\kappa_0}\delta_1\end{pmatrix} \\ & + \frac{\beta}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\begin{pmatrix}0\\ \xi_1\end{pmatrix} + \frac{\beta}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\sqrt{\frac{\nu_0 N}{\omega_0\kappa_0}}\,\zeta_1\begin{pmatrix}0\\ \delta_1\end{pmatrix}. \end{aligned} \tag{27}$$

A cleaner way to write (27) is to use the block-normal decomposition of $\bar{z}_0$. Define

$$a_0 = \frac{\bar{z}_{01}}{\sqrt{\omega_0}}, \qquad r_0 = \frac{\delta_1}{\sqrt{\omega_0}}.$$

Conditional on ω_0,

$$a_0 \sim \mathcal{N}\left(0, \frac{1}{N\omega_0}\right), \qquad r_0 \sim \mathcal{N}_{J-1}\left(0, \frac{\bar{S}_1}{N\omega_0}\right), \qquad a_0 \perp r_0,$$

and

$$\kappa_0 = \nu_0 + N a_0^2.$$

Then the extracted model becomes the simpler block system

$$x_1 = \beta + a_0, \tag{28}$$

$$\begin{aligned} x_{-1} &= \mu_1 x_1 + \frac{\beta}{\sqrt{\omega_1 \kappa_0 (\nu_0 + 4)}} \xi_1 \\ &\quad + \left[1 + \frac{\beta N a_0}{\kappa_0} + \frac{\beta}{\sqrt{\omega_1 \kappa_0 (\nu_0+4)}} \sqrt{\frac{\nu_0 N}{\kappa_0}} \zeta_1\right] r_0. \end{aligned} \tag{29}$$

This is the nicest form of the last equation: the fixed-noise part has been split into a scalar noise a_0 in the active coordinate and an independent conditional residual r_0 in the null coordinates.

This form also shows what can be integrated out. Since $x_1 = \beta + a_0$, once we condition on (x_1, β) we may replace

$$a_0 = x_1 - \beta, \qquad \kappa_0 = \nu_0 + N(x_1 - \beta)^2.$$

Integrating out r_0 and ξ_1 conditional on $(x_1, \beta, \omega_0, \omega_1, \zeta_1)$ gives

$$x_{-1} \mid x_1, \beta, \omega_0, \omega_1, \zeta_1 \sim \mathcal{N}_{J-1}\left(\mu_1 x_1,\ V_{-1}(x_1, \beta, \omega_0, \omega_1, \zeta_1) \bar{S}_1\right), \tag{30}$$

where

$$\begin{aligned} V_{-1} &= \frac{1}{N\omega_0}\left[1 + \frac{\beta N (x_1 - \beta)}{\kappa_0} + \frac{\beta}{\sqrt{\omega_1 \kappa_0 (\nu_0+4)}} \sqrt{\frac{\nu_0 N}{\kappa_0}} \zeta_1\right]^2 \\ &\quad + \frac{\beta^2 \nu_0}{\omega_1 \kappa_0 (\nu_0 + 4)}. \end{aligned} \tag{31}$$

Together with

$$x_1 \mid \beta, \omega_0 \sim \mathcal{N}\left(\beta, \frac{1}{N\omega_0}\right),$$

this integrates out the Gaussian pieces of $\bar{z}_0$ in closed form. What does not seem to simplify further is the integral over ζ_1: the variance in (31) is quadratic in ζ_1, so integrating $\zeta_1 \sim \mathcal{N}(0,1)$ gives a one-dimensional normal scale mixture rather than another normal distribution. Thus $\bar{z}_0$ can be integrated out conditionally, but not in a way that keeps the final model as a simple fixed-noise Gaussian likelihood.

3.1 Clean equations

$$\begin{cases} x = \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix} + z_0 \\ z_0 \mid R \sim \mathcal{N}(0, R/N) \\ z_1 = \dfrac{R_{-1,1}}{R_{11}} \\ R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1) \end{cases} \tag{32}$$

with

$$z_0 = \frac{\bar{z}_0}{\sqrt{\omega_0}}, \quad \bar{z}_0 \sim \mathcal{N}\left(0, \frac{\bar{R}}{N}\right), \quad \omega_0 \sim \text{Gamma}\left(\frac{\nu_0+2}{2}, \frac{\nu_0}{2}\right). \tag{33}$$

Partition

$$\bar{z}_0 = \begin{pmatrix} \bar{z}_{01} \\ \bar{z}_{0,-1} \end{pmatrix}, \qquad \delta_1 = \bar{z}_{0,-1} - \bar{z}_{01}\mu_1, \qquad \kappa_0 = \nu_0 + \frac{N}{\omega_0}\bar{z}_{01}^2.$$

Then, with

$$\omega_1 \sim \text{Gamma}\left(\frac{\nu_0+4}{2}, \frac{\nu_0+4}{2}\right), \qquad \xi_1 \sim \mathcal{N}_{J-1}(0, \nu_0\bar{S}_1), \qquad \zeta_1 \sim \mathcal{N}(0,1),$$

we can write the relaxed LD column cleanly as

$$\boxed{z_1 = \mu_1 + \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\xi_1 + \left[\frac{N\bar{z}_{01}}{\omega_0\kappa_0} + \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\sqrt{\frac{\nu_0 N}{\omega_0\kappa_0}}\,\zeta_1\right]\delta_1.} \tag{34}$$

Equivalently, conditional on $(\bar{z}_0, \omega_0, \omega_1, \zeta_1)$, the only remaining randomness in z_1 is the baseline Gaussian vector ξ_1.

Intuition. The last display makes it easy to see why the marginal scale of z_1 should be close to $\bar{S}_1/\nu_0$. The key observation is that, under $\bar{z}_0 \sim \mathcal{N}(0, \bar{R}/N)$,

$$\bar{z}_{01} \sim \mathcal{N}\left(0, \frac{1}{N}\right), \qquad \delta_1 = \bar{z}_{0,-1} - \bar{z}_{01}\mu_1 \sim \mathcal{N}_{J-1}\left(0, \frac{\bar{S}_1}{N}\right), \qquad \bar{z}_{01} \perp \delta_1.$$

Thus both ξ_1 and δ_1 have covariance matrices proportional to the same matrix $\bar{S}_1$:

$$\xi_1 \sim \mathcal{N}_{J-1}(0, \nu_0\bar{S}_1), \qquad \delta_1 \sim \mathcal{N}_{J-1}\left(0, \frac{\bar{S}_1}{N}\right).$$

For intuition, condition on the scalar variables $(\bar{z}_{01}, \omega_0, \omega_1)$, but average over $(\xi_1, \delta_1, \zeta_1)$. Write

$$A = \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}, \qquad B = \frac{N\bar{z}_{01}}{\omega_0\kappa_0}, \qquad C = A\sqrt{\frac{\nu_0 N}{\omega_0\kappa_0}}.$$

Then

$$z_1 - \mu_1 = A\xi_1 + (B + C\zeta_1)\delta_1,$$

so, using independence and $\mathbb{E}(\zeta_1^2) = 1$,

$$\begin{aligned} \text{Var}(z_1 \mid \bar{z}_{01}, \omega_0, \omega_1) &= \left\{A^2\nu_0 + \frac{B^2 + C^2}{N}\right\}\bar{S}_1 \\ &= \left\{\frac{\nu_0}{\omega_1\kappa_0(\nu_0+4)} + \frac{N\bar{z}_{01}^2}{\omega_0^2\kappa_0^2} + \frac{\nu_0}{\omega_0\omega_1\kappa_0^2(\nu_0+4)}\right\}\bar{S}_1. \end{aligned} \tag{35}$$

Now typically $\omega_0 \approx 1$, $\omega_1 \approx 1$, and $N\bar{z}_{01}^2 = O_p(1)$, so

$$\kappa_0 = \nu_0 + \frac{N}{\omega_0}\bar{z}_{01}^2 = \nu_0 + O_p(1).$$

Therefore the first term in braces in (35) is

$$\frac{\nu_0}{\omega_1\kappa_0(\nu_0+4)} \approx \frac{1}{\nu_0},$$

whereas the two extra terms are of order $O_p(\nu_0^{-2})$. Hence, to first order,

$$\mathrm{Var}(z_1) \approx \frac{\bar{S}_1}{\nu_0}.$$

This matches the original inverse-Wishart block intuition $z_1 \mid S_1 \sim \mathcal{N}(\mu_1, S_1/\nu_0)$ with S_1 concentrated near $\bar{S}_1$.

This decomposition also clarifies how model B differs from SER-relax. The term

$$\mu_1 + \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\xi_1$$

is the SER-relax-like part: it is centered at the reference LD column μ_1, and because

$$\frac{\xi_1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}} \approx \mathcal{N}_{J-1}\left(0, \frac{\bar{S}_1}{\nu_0}\right),$$

it behaves like a relaxed LD column with variance approximately $\bar{S}_1/\nu_0$. The remaining part,

$$\left[\frac{N\bar{z}_{01}}{\omega_0\kappa_0} + \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\sqrt{\frac{\nu_0 N}{\omega_0\kappa_0}}\,\zeta_1\right]\delta_1,$$

is the additional uncertainty carried by model B. It is tied to the realized noise vector $\bar{z}_0$ through the conditional residual $\delta_1 = \bar{z}_{0,-1} - \bar{z}_{01}\mu_1$. This extra term is small to first order when ν_0 is large, but it is precisely the part that couples the relaxed LD column to the noise realization in model B and is absent from a plain SER-relax-style column perturbation.

Calculating PIP from this hierarchical representation. The hierarchy above is not exactly an equivalent representation of Model B, because the active-coordinate noise still contains the random diagonal of R. It is instead closer to the original inverse-Wishart model with a random active diagonal. Written for $\gamma = 1$, the hierarchy is

$$\begin{cases} x = \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix} + z_0 \\ z_0 = \dfrac{\bar{z}_0}{\sqrt{\omega_0}} \\ \bar{z}_0 \sim \mathcal{N}\left(0, \frac{\bar{R}}{N}\right) \\ \omega_0 \sim \mathrm{Gamma}\left(\frac{\nu_0+2}{2}, \frac{\nu_0}{2}\right) \\ z_1 = \mu_1 + \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\xi_1 + \left[\frac{N\bar{z}_{01}}{\omega_0\kappa_0} + \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\sqrt{\frac{\nu_0 N}{\omega_0\kappa_0}}\,\zeta_1\right]\delta_1 \\ \omega_1 \sim \mathrm{Gamma}\left(\frac{\nu_0+4}{2}, \frac{\nu_0+4}{2}\right) \\ \xi_1 \sim \mathcal{N}_{J-1}(0, \nu_0\bar{S}_1) \\ \zeta_1 \sim \mathcal{N}(0,1) \end{cases} \tag{36}$$

where

$$\bar{z}_0 = \begin{pmatrix} \bar{z}_{01} \\ \bar{z}_{0,-1} \end{pmatrix}, \qquad \delta_1 = \bar{z}_{0,-1} - \bar{z}_{01}\mu_1, \qquad \kappa_0 = \nu_0 + \frac{N}{\omega_0}\bar{z}_{01}^2.$$

My goal now is to find a sequence of marginalization so we get back to the appropriate PIP. Note that $\bar{z}_{01}$ and δ_1 are independent, and they are independent with ξ_1.

3.2 Adding the random diagonal to the latent-noise representation

The preceding issue suggests the right correction. If the noise is generated from the random in-sample LD matrix, then the mean should use the same random active column. For $\gamma = 1$, the analogue of (36) is

$$\begin{cases} x = \beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix} + z_0 \\ z_0 = \dfrac{\bar{z}_0}{\sqrt{\omega_0}} \\ \bar{z}_0 \sim \mathcal{N}\left(0, \dfrac{\bar{R}}{N}\right) \\ \omega_0 \sim \text{Gamma}\left(\dfrac{\nu_0+2}{2}, \dfrac{\nu_0}{2}\right) \\ d_1 \mid \bar{z}_0, \omega_0 \sim \text{IG}\left(\dfrac{\nu_0+3}{2}, \dfrac{\kappa_0}{2}\right) \\ z_1 = \mu_1 + \dfrac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\xi_1 + \left[\dfrac{N\bar{z}_{01}}{\omega_0\kappa_0} + \dfrac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}\sqrt{\dfrac{\nu_0 N}{\omega_0\kappa_0}}\,\zeta_1\right]\delta_1 \\ \omega_1 \sim \text{Gamma}\left(\dfrac{\nu_0+4}{2}, \dfrac{\nu_0+4}{2}\right) \\ \xi_1 \sim \mathcal{N}_{J-1}(0, \nu_0\bar{S}_1) \\ \zeta_1 \sim \mathcal{N}(0,1), \end{cases} \tag{37}$$

where

$$\bar{z}_0 = \begin{pmatrix} \bar{z}_{01} \\ \bar{z}_{0,-1} \end{pmatrix}, \qquad \delta_1 = \bar{z}_{0,-1} - \bar{z}_{01}\mu_1, \qquad \kappa_0 = \nu_0 + \frac{N}{\omega_0}\bar{z}_{01}^2.$$

This is the same posterior-reversal construction as before, but now it also keeps the random diagonal d_1. The conditional inverse-gamma distribution for d_1 is part of this hierarchy, and its scale is determined by the active coordinate of the realized noise through $\kappa_0 = \nu_0 + N\bar{z}_{01}^2/\omega_0$.

Directly integrating out ξ_1 and δ_1. This is a reasonable first marginalization. Write

$$A = \frac{1}{\sqrt{\omega_1\kappa_0(\nu_0+4)}}, \qquad B = \frac{N\bar{z}_{01}}{\omega_0\kappa_0}, \qquad C = A\sqrt{\frac{\nu_0 N}{\omega_0\kappa_0}},$$

so that

$$z_1 = \mu_1 + A\xi_1 + (B + C\zeta_1)\delta_1.$$

The observation equation in (37) gives

$$x_1 = \beta d_1 + \frac{\bar{z}_{01}}{\sqrt{\omega_0}},$$

and

$$\begin{aligned} x_{-1} &= \beta d_1 z_1 + \frac{\bar{z}_{0,-1}}{\sqrt{\omega_0}} \\ &= \mu_1 x_1 + \beta d_1 A \xi_1 + \left\{ \frac{1}{\sqrt{\omega_0}} + \beta d_1 (B + C\zeta_1) \right\} \delta_1. \end{aligned}$$

Because

$$\xi_1 \sim \mathcal{N}_{J-1}(0, \nu_0 \bar{S}_1), \qquad \delta_1 \sim \mathcal{N}_{J-1}\left(0, \frac{\bar{S}_1}{N}\right), \qquad \xi_1 \perp \delta_1,$$

integrating out (ξ_1, δ_1) conditional on the scalar variables gives

$$x_{-1} \mid x_1, \beta, d_1, \bar{z}_{01}, \omega_0, \omega_1, \zeta_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(\mu_1 x_1, V_1 \bar{S}_1\right), \tag{38}$$

where

$$\begin{aligned} V_1 = &\frac{\beta^2 d_1^2 \nu_0}{\omega_1 \kappa_0 (\nu_0 + 4)} \\ &+ \frac{1}{N} \left\{ \frac{1}{\sqrt{\omega_0}} + \beta d_1 \frac{N \bar{z}_{01}}{\omega_0 \kappa_0} + \beta d_1 \frac{1}{\sqrt{\omega_1 \kappa_0 (\nu_0 + 4)}} \sqrt{\frac{\nu_0 N}{\omega_0 \kappa_0}}\, \zeta_1 \right\}^2. \end{aligned} \tag{39}$$

Equivalently, after using the active-coordinate constraint

$$a_1 := x_1 - \beta d_1 = \frac{\bar{z}_{01}}{\sqrt{\omega_0}}, \qquad \kappa_0 = \nu_0 + N a_1^2,$$

the variance factor becomes

$$\begin{aligned} V_1 = &\frac{1}{N\omega_0} \left[1 + \frac{\beta d_1 N a_1}{\kappa_0} + \frac{\beta d_1}{\kappa_0} \sqrt{\frac{\nu_0 N}{\omega_1 (\nu_0 + 4)}}\, \zeta_1 \right]^2 \\ &+ \frac{\beta^2 d_1^2 \nu_0}{\omega_1 \kappa_0 (\nu_0 + 4)}. \end{aligned} \tag{40}$$

Thus this first marginalization does stay entirely inside (37). It leaves a scalar normal-scale mixture: the remaining integral over $(\beta, d_1, \omega_0, \omega_1, \zeta_1)$ contains

$$|\bar{S}_1|^{-1/2} V_1^{-(J-1)/2} \exp\left\{ -\frac{(x_{-1} - \mu_1 x_1)^\top \bar{S}_1^{-1} (x_{-1} - \mu_1 x_1)}{2V_1} \right\}.$$

This expression already shows that the null coordinates enter through one quadratic form. This leads to a direct PIP expression using only the latent variables in (37).

Direct PIP expression. For a general active SNP j, define

$$\mu_j = \bar{R}_{-j,j}, \qquad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top, \qquad r_j = x_{-j} - x_j \mu_j,$$

and

$$s_j = r_j^\top \bar{S}_j^{-1} r_j.$$

Using the block inverse formula for $\bar{R}$,

$$s_j = x^\top \bar{R}^{-1} x - x_j^2.$$

Let $p = J - 1$, and for scalar integration variables $(\beta, d, \omega_0, \omega_1, \zeta)$ define

$$a_j = x_j - \beta d, \qquad \kappa_j = \nu_0 + N a_j^2,$$

and

$$V_j(\beta, d, \omega_0, \omega_1, \zeta) = \frac{1}{N\omega_0} \left[1 + \frac{\beta d N a_j}{\kappa_j} + \frac{\beta d}{\kappa_j} \sqrt{\frac{\nu_0 N}{\omega_1(\nu_0 + 4)}}\, \zeta \right]^2 + \frac{\beta^2 d^2 \nu_0}{\omega_1 \kappa_j (\nu_0 + 4)}. \tag{41}$$

Then the marginal likelihood under $\gamma = j$ can be written, up to constants that do not depend on j, as

$$\begin{aligned} m_j(x) = |\bar{S}_j|^{-1/2} \int & \mathcal{N}(\beta \mid 0, \sigma_0^2)\, \mathrm{Gamma}\left(\omega_0; \frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ & \times \omega_0^{1/2} \mathcal{N}\left(\sqrt{\omega_0} a_j \mid 0, \frac{1}{N}\right) \mathrm{IG}\left(d; \frac{\nu_0 + 3}{2}, \frac{\kappa_j}{2}\right) \\ & \times \mathrm{Gamma}\left(\omega_1; \frac{\nu_0 + 4}{2}, \frac{\nu_0 + 4}{2}\right) \mathcal{N}(\zeta \mid 0, 1) \\ & \times V_j(\beta, d, \omega_0, \omega_1, \zeta)^{-p/2} \exp\left\{ -\frac{s_j}{2 V_j(\beta, d, \omega_0, \omega_1, \zeta)} \right\} d\beta\, dd\, d\omega_0\, d\omega_1\, d\zeta. \end{aligned} \tag{42}$$

The factor $\omega_0^{1/2}$ is the Jacobian from substituting $\bar{z}_{0j} = \sqrt{\omega_0}(x_j - \beta d)$ using the active-coordinate equation. Finally,

$$\boxed{\Pr(\gamma = j \mid x) = \frac{m_j(x)}{\sum_{k=1}^{J} m_k(x)}.} \tag{43}$$

This is not as simple as the earlier one-dimensional integral, but it is derived directly from (37). It also shows the same qualitative irrelevance property: since $|\bar{S}_j| = |\bar{R}|$ when $\bar{R}_{jj} = 1$, and since $s_j = x^\top \bar{R}^{-1} x - x_j^2$, all SNP-specific quantities in (42) are functions of x_j and the common scalar $x^\top \bar{R}^{-1} x$. By the symmetry $(x_j, \beta, \zeta) \mapsto (-x_j, -\beta, -\zeta)$, the weight is an even function of x_j. Thus, for fixed hyperparameters and fixed common $x^\top \bar{R}^{-1} x$, the direct latent-noise PIP weight depends on SNP j through x_j^2.